



Elastomeric fluorinated polyurethane coatings for nontoxic fouling control

Robert F Brady & Carl L Aronson

To cite this article: Robert F Brady & Carl L Aronson (2003) Elastomeric fluorinated polyurethane coatings for nontoxic fouling control, , 19:S1, 59-62, DOI: [10.1080/0892701031000061741](https://doi.org/10.1080/0892701031000061741)

To link to this article: <https://doi.org/10.1080/0892701031000061741>



Published online: 09 Sep 2010.



Submit your article to this journal [↗](#)



Article views: 272



View related articles [↗](#)



Citing articles: 3 View citing articles [↗](#)

Short Communication

Elastomeric Fluorinated Polyurethane Coatings for Nontoxic Fouling Control

ROBERT F BRADY JR* and CARL L ARONSON†

Chemistry Division, Naval Research Laboratory, Washington, DC 20375-5342, USA

(Received 23 July 2002; in final form 30 October 2002)

Nontoxic antifouling coatings have been investigated for many years as possible successors to toxic antifouling paints. Polymers containing fluorine or silicone have been tested and each has been shown to be partially effective for different reasons. This paper describes a new coating which combines the best features of fluorinated and silicone coatings and is non-toxic. Four fluorinated elastomers were prepared and tested for fouling resistance during a full fouling season. The surface energy and mechanical properties of each polymer were measured and correlated to fouling performance. One of the elastomers was shown to foul slowly, clean easily, be durable in the marine environment and organisms bonded to it only weakly. The surface energy, elastic modulus, and thickness of the elastomer may be varied as desired over wide ranges to meet differing performance requirements.

Keywords: adhesion; elastic modulus; elastomeric coating; marine fouling; fracture; fluoropolymer; peel; surface energy

INTRODUCTION

Throughout history sailors have used poisons to keep marine organisms off of the hulls of their ships (Callow, 1990; Pain, 1999). The technology has advanced from the copper sheathing first used on the hulls of British ships in 1779 to the tin-based paints in use today. However, regulations issued recently by the Marine Environment Protection Committee of the International Maritime Organization, a division of the United Nations, stipulate that tin-based antifouling coatings are not to be

applied to ships after January 2003, and are to be removed from all ships by January 2008 (Reisch, 2001). Thus the search for a nontoxic replacement for current hull coatings has sudden urgency.

Formulation (Brady, 2000) and testing (Lewis *et al.*, 1997; Jones-Meehan *et al.*, 1999; Kavanagh *et al.*, 2001) of environmentally-friendly, nontoxic antifouling coatings have been pursued in laboratories around the world for some 25 years. Although conventional theories predict that adhesion is proportional to the surface free energy of the substrate, the converse has been shown to be true (Zhang Newby *et al.*, 1995; Zhang Newby & Chaudhury, 1997). Extended studies have also shown that the fracture mechanics of the joint between the coating and the fouling organism is a critical factor in determining fouling release (Kohl & Singer, 1999; Brady & Singer, 2000).

Two materials have shown the greatest potential for success as nontoxic antifouling coatings. Fluorinated coatings are tough, have low surface energy and resist fouling, but ultimately form a tight bond to fouling. The bond between fouling and a fluorinated coating fails in shear, a relatively high-energy process, without damage to the coating. Silicone coatings, on the other hand, are soft, relatively weak, and easily damaged or torn by marine debris. Because their surface energy is higher than that of fluoropolymers, they foul more rapidly and form a somewhat stronger bond with fouling organisms.

*Corresponding author; present address: 706 Hope Lane, Gaithersburg, MD 20878-1883 USA ; e-mail: SharonandBobB@aol.com

†present address: Department of Science and Mathematics, Kettering University, 1700 West Third Avenue, Flint, MI 48504, USA; e-mail: caronson@kettering.edu

TABLE I Properties that contribute to the effectiveness of nontoxic antifouling coatings

All coatings	Fluorinated coatings	Silicone coatings
Smooth surface, to avoid mechanical interlocking and to provide a sharp, easily-cleaved interface	A surface composed exclusively of fluorinated groups	A flexible, linear silicon-oxygen backbone
Absence of heteroatoms, ions and dipoles on the surface, to avoid polar interactions with marine adhesives	Sufficient fluorine in the bulk of the coating to effectively control the organization of fluorine at the surface	Groups on the backbone that reduce surface energy while maintaining backbone mobility
Chemical and physical stability in the marine environment	Dipoles such as $\text{CF}_2\text{-CH}_2$ concealed well beneath the surface	Low elastic modulus consistent with physical toughness and durability
	A crosslinked surface which holds fluorinated groups in place and resists infiltration of marine adhesives	Hydrolytic stability to avoid loss of mass and development of surface roughness
		A thickness that favors fracture of the interface by peel rather than by shear

Application of force to the joint deforms the rubbery silicone and the resin peels away from the marine adhesive in a process that is slower but requires less energy than shear. A summary of the properties that contribute to the effectiveness of nontoxic antifouling coatings are given in Table I.

In this paper the production of a coating that combines the best features of fluorinated and silicone antifouling coatings is described. For the first time, a tough fluorinated elastomeric coating has been made that tolerates no more than a weak or imperfect joint between the foulant and itself, a joint which is predisposed to early and easy failure. The elastomer has low surface energy, resists the attachment of fouling and also has a low elastic modulus, which permits the joint between foulant and coating to fail in peel. In addition, it has the ability to be applied in a thickness which favors failure of the fouling-coating joint by peel.

MATERIALS AND METHODS

Polyols and polyisocyanates were obtained from the manufacturers and were used as received. The polyols included polypropylene glycol (PPG) (Aldrich, equivalent weight 212), a fluorinated ether-vinyl ether (FVE) supplied as a 50% wt/wt solution in *n*-butyl acetate (Asahi Glass Incorporated, eqt wt of solution 906) (Handforth, 1990), and a polyol made from hexafluoroacetone (HFA) and supplied as a 70% wt/wt solution in *n*-butyl acetate/xylene (21st Century Coatings Incorporated, eqt wt of solution 535) (Field, 1976). The polyisocyanates included polypropylene glycol endcapped with toluene diisocyanate (PDI) (Aldrich, 8.4 % free isocyanate, eqt wt 500), and a perfluoropolyether (FEI) of the Fomblin® type endcapped with isophorone diisocyanate (70%) and toluene diisocyanate (30%) supplied as a 60% wt./wt. solution in *n*-butyl

acetate (Fluorobase Z 1072H, Ausimont Chemical Company, eqt wt of solution 1200). The formulations also contained the triol trimethylolpropane propoxylate (Aldrich, MW 308, eqt wt 103) as a crosslinker and dibutyltin dilaurate (MW 631) as a catalyst.

Elastomers were prepared by mixing a base component and a curing agent. Unreactive oils and other ingredients that would slowly leach from the coating were not used in the formulation. The base component contained a diol, a triol, and a catalyst, and the curing agent contained a diisocyanate. For example, a base component was made by mixing a fluorinated polyol (HFA, 6.42 g), trimethylolpropane propoxylate (0.41 g), and dibutyltin dilaurate (0.04 g). The curing agent was polypropylene glycol endcapped with toluene diisocyanate (PPG, 9.60 g), used as supplied. All combinations of polyol and polyisocyanate were used with triol and catalyst. The stoichiometric equivalence of isocyanate to hydroxyl was 1.1, 1.2 or 1.3.

The base and curing agent were mixed thoroughly for less than 5 min, poured into a clean Petri dish to form a layer 6.0 ± 0.2 mm thick, and allowed to cure at room temperature for 48 h. The elastomer was removed from the dish and allowed to cure in air for an additional 7 d before testing. Tests were conducted on the side of the elastomer which formed against the glass.

Contact angle measurements were made with an AST 2500 Video Contact Angle System as described by Bonafede *et al.*, (1999). Contact angles were determined from 2 μ l sessile drops of three test liquids, *viz.* triple-distilled water, 99% methylene iodide, and 99+% spectrophotometric grade formamide. Drops were applied separately to the elastomer, and the contact angle between the drop and the surface was measured approximately 5–10 s after the drop was applied. The measured contact angles and the surface tensions of the test liquids were used to calculate surface energy by

the geometric mean method. The reported surface energies are the mean of 16–20 measurements.

Seawater immersion tests were conducted in the Chesapeake Bay at Deale, MD during the summer 2000 fouling season. Elastomers were affixed to polymethyl methacrylate panels with epoxy cement and immersed to a depth of 1 to 4 feet (depending on tides). Fouling was evaluated at approximately 2 month intervals using assessment procedures described elsewhere (Bonafede & Brady, 1998). At each inspection the extent of fouling organisms, ease of cleaning, residual biofouling, and integrity of (damage to) the cleaned elastomer were judged and rated on a scale of 0 (best) to 5 (worst).

Hardness and elastic modulus were measured with a Fisherscope H-100 Microindenter which senses both load and displacement. The applied load on a Vickers indenter (four-sided pyramid) was increased incrementally in 30 one second steps to three maximum load levels, *viz.* 10, 30 and 100 mN. WinUMIS 2 software was used for analysis of the data.

RESULTS AND DISCUSSION

Six polymers were prepared from three polyols and two polyisocyanates. Two of these, HFA-FDI and PPG-PDI were not elastomers and were excluded from further study. For the remaining four elastomers, the experimentally-determined dispersive component γ_d , polar component γ_p , and total surface energy γ_T as well as the fluorine content (calculated from the fluorine contents of the ingredients) are shown in Table II. Surface energies of the fluorinated elastomers were well below corresponding values of γ_T for commodity polymers such as polystyrene ($\gamma_T = 40.0 \text{ dynes cm}^{-1}$)

TABLE III Immersion testing of fluorinated polyurethane elastomers

NCO/OH ratio	Fouling	Ease of cleaning	Residual fouling	Integrity
HFA-PDI				
1.1	3	1	0	0
1.2	3	1	0	0
1.3	3	1	0	0
FVE-FDI				
1.1	3	2	2	0
1.2	3	3	2	0
1.3	3	1	0	0
FVE-PDI				
1.1	4	1	0	0
1.2	3	3	0	0
1.3	4	1	0	0
PPG-FDI				
1.1	3	2	0	0
1.2	3	2	2	0
1.3	2	3	0	3

and polyethylene ($\gamma_T = 33 \text{ dynes cm}^{-1}$), indicating that the fluorine atoms were aggregated at the surface of the material.

Fouling performance was evaluated monthly from May to November during the fouling season. Fouling was heaviest in August, after 3 months immersion. The data in Table III are for the August inspection and are representative of the performance during the entire fouling season. All materials accumulated barnacles, bryozoans and algae, and the extent of fouling was moderate.

The elastic and plastic properties of the four elastomers are given in Table IV. The value for universal hardness H_U includes both plastic and elastic hardness. The percentage recovery was calculated by dividing the elastic energy recovered by the total energy expended in indenting the material.

TABLE II Fluorine content and surface energy of elastomers

Composition	NCO/OH ratio	Fluorine percent	γ_d mJ m ⁻²	γ_p mJ m ⁻²	γ_T mJ m ⁻²
HFA-PDI					
	1.1	10	21.8	3.4	25.2
	1.2	10	24.9	3.2	27.1
	1.3	9	24.9	3.2	27.1
FVE-FDI					
	1.1	37	5.2	3.9	9.1
	1.2	37	9.5	1.6	11.1
	1.3	38	4.2	6.5	10.7
FVE-PDI					
	1.1	10	8.5	17.4	25.9
	1.2	9	10.3	19.2	29.5
	1.3	9	8.9	20.8	29.7
PPG-FDI					
	1.1	35	7.2	1.8	9.0
	1.2	36	4.3	4.8	9.1
	1.3	36	6.8	2.5	9.3

TABLE IV Elastic and plastic properties of fluorinated polyurethane elastomers

NCO/OH ratio	H_U MPa	E MPa	Recovery %
HFA-PDA-PDI			
1.2	0.36	4.8	72
1.2	0.49	6.2	75
1.3	0.31	3.9	76
FVE-FDI			
1.1	5.2	173	36
2.1	59	1420	36
2.1	60	1590	35
FVE-PDI			
1.1	0.19	35	52
1.2	0.76	18	56
1.3	3.7	50	35
PPG-FDI			
1.1	0.77	10.2	74
1.2	0.63	9.7	64
1.3	0.94	1.8	51

Material FVE-FDI had a high content of fluorine and a low surface energy. However, it was the hardest of these elastomers, had the highest elastic modulus, and the least recovery of elastic energy after indentation. It was a tough elastomer that also retained some of the attributes of the glassy fluorinated coatings. Although it fouled slowly, it was not sufficiently elastomeric to improve its cleaning properties.

Elastomer PPG-FDI also had a high fluorine content and a low surface energy. It was about as hard as FVE-PDI, but had a much smaller Young's modulus and a greater recovery of elastic energy. It also fouled slowly and was difficult to clean, and it was the only elastomer damaged by immersion and cleaning. It was significantly more elastomeric than FVE-FDI, but it was the most difficult to clean and retained more staining than other materials. Both of these materials, FVE-FDI and PPG-FDI, support the determination (Brady, 2000) that high fluorine content and low surface energy are in themselves not sufficient to impart ease of cleaning.

Elastomer FVE-PDI had a low content of fluorine and the highest surface energy of the four polymers in this study. This material was unlike the others in that the polar component of its surface energy exceeded the dispersive component, and this difference was large. The universal hardness of this material was not great, but its elastic modulus was moderately high. It accumulated more fouling than other materials, but released it easily and suffered no residual staining or damage.

Elastomer HFA-PDI also had a low content of fluorine and a moderate surface energy, almost as high as FVE-PDI. As in most of the other materials, the dispersive component of its surface energy was greater than the polar component, but the seven- to eight-fold magnitude of the difference has not been observed in similar materials that have been studied. It was the softest material in this series, and recovered the greatest proportion of elastic energy. It fouled, but was no worse than the other materials in this study, and it cleaned quite easily without residual staining or damage.

The surface energies of HFA-PDI and FVE-PDI were the highest of all materials studied here, but by comparison to other organic polymers (polyethylene, 33.7 mJ m^{-2} , polystyrene, 40.0 mJ m^{-2} ,

nylon 66, 45.9 mJ m^{-2}), the surface energies were moderately low.

CONCLUSIONS

The fluorinated elastomer HFA-PDI fouled slowly and marine organisms were bonded weakly; it was easily cleaned and was durable in the marine environment. The surface energy of this elastomer was higher than many fluoropolymers but was sufficient to retard the rate of accumulation and strength of attachment of marine organisms.

Acknowledgement

The authors thank Robert N Bolster for the mechanical property data and Irwin Singer for insightful technical discussions.

References

- Bonafede S J, Brady R F Jr (1998) Compositional effects on the fouling resistance of fluorourethane coatings. *Surf Coat Intl* **81**: 181–185
- Bonafede S J, Pawlikowski G T, Brady R F Jr (1999) Macrofouling of hydrocarbon polymers which contain variable proportions of crystallinity. *Biofouling* **14**: 1–14
- Brady R F (2000) Clean hulls without poisons: devising and testing nontoxic marine coatings. *J Coatings Tech* **72**: 44–56
- Brady R F Jr, Singer I L (2000) Mechanical factors favoring release from fouling release coatings. *Biofouling* **15**: 73–81
- Callow M (1990) Ship fouling: problems and solutions. *Chem Ind (Lond)* **May**: 123–127
- Field D E (1976) Fluorinated polyurethane and polyepoxy coatings. *J Coatings Tech* **48**: 43–47
- Handforth V (1990) Properties and applications of novel fluoropolymer resins. *J Oil Colour Chem Assoc* **73**: 145–148
- Jones-Meehan J, Cella J, Montemarano J A, Swain, G W, Wiebe D, Meyer A, Baier, R E (1999) Advanced nontoxic fouling release coatings. Naval Research Laboratory Report NRL/PU/6110-99-388
- Kavanagh C J, Schultz M P, Swain G W, Stein J, Truby K, Darkangelo Wood C (2001) Variation in adhesion strength of *Balanus eburneus*, *Crassostrea virginica* and *Hydroides dianthus* to fouling release coatings. *Biofouling* **17**: 155–167
- Kohl J R, Singer I L (1999) Pull-off behavior of epoxy bonded to silicone duplex coatings. *Prog Org Coatings* **36**: 15–20
- Lewis J A, Foster T, and Brady R F Jr (1997) Environmentally friendly flexible fouling release or prevention coatings. Final Report, TTCP MAT Group Assignment MAT TP-6-O9
- Pain S (1999) How to rule the waves. *New Sci* **162**: 54–55
- Reisch M (2001) Maritime dilemma. *Chem Eng News* **79**: 16
- Zhang Newby B-M, Chaudhury M K, (1997) Effect of interfacial slippage on viscoelastic adhesion. *Langmuir* **13**: 1805–1809
- Zhang Newby B-M, Chaudhury M K, Brown H R (1995) Macroscopic evidence of the effect of interfacial slippage on adhesion. *Science* **269**: 1407–1409